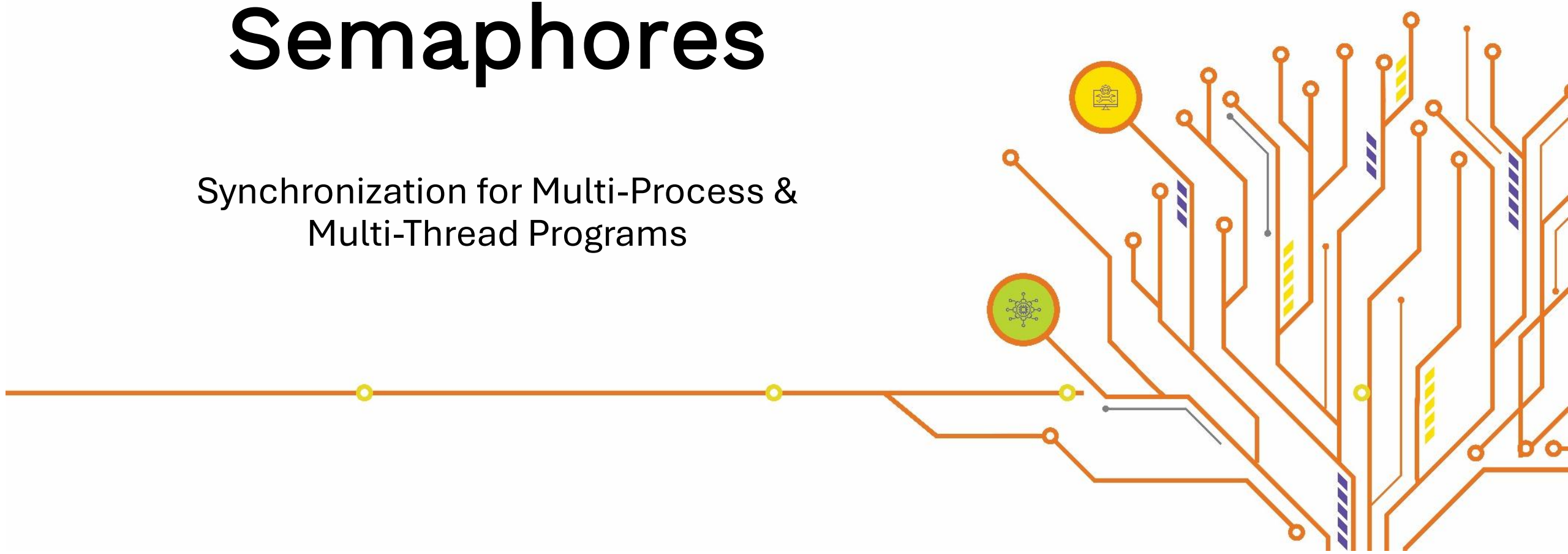
KPIT | APEX LAB

POSIX Semaphores

Synchronization for Multi-Process & Multi-Thread Programs



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What is a Semaphore?

A synchronization primitive

- Manages access to shared resources
- Based on a counter
- Processes/threads wait when counter = 0
- Used for:
  - mutual exclusion
  - producer–consumer
  - signaling events
  - limiting access to N resources

Counter ↓ as tasks enter
↑ as they leave

Types of POSIX Semaphores

Unnamed Semaphores	Named Semaphores
• Stored in memory • Used between threads or processes (with shared memory) • Created using sem_init()	• Identified by a name (e.g., /semApex) • Work across unrelated processes • Created using sem_open()

Semaphore Operations

- sem_wait()
  - Decrements counter
  - Blocks if counter = 0
- sem_post()
  - Increments counter
  - Wakes waiting process/thread
- sem_getvalue()
  - Read current count (debug/testing)

Unnamed Semaphore (sem_init)

- Prototype:
  - int sem_init(sem_t *sem, int pshared, unsigned int value);
  - Fields:
    - pshared = 0 → between threads
    - pshared = 1 → between processes (in shared memory)
    - value → initial count
- Example:
  - sem_t lock;sem_init(&lock, 0, 1); // binary semaphore

Named Semaphores (sem_open)

- Prototype:
  - sem_t *sem_open(const char *name, int oflag, mode_t mode, unsigned int value);
- Usage:
  - sem_t *s = sem_open("/apexsem", O_CREAT, 0666, 1);
- Notes:
  - Name must begin with /
  - Exists until removed using sem_unlink()
  - Works between unrelated processes

Binary vs Counting Semaphores

Binary Semaphores	Counting Semaphores
• Value = 1 • Used like a mutex • Ensures exclusive access	• Value = N • Allows up to N simultaneous accesses • Used for resource pools, producer–consumer

Example (Binary Semaphore — Mutual Exclusion)

```
sem_t lock;

void *worker(void *arg) {
   sem_wait(&lock);
   // critical section
   sem_post(&lock);
}
```

- Use case:
  - Protect shared variable
  - Protect file writes / hardware access
  - Avoid race conditions

Semaphore Lifecycle

Create: sem_init() → Use: sem_wait() / sem_post() → Destroy: sem_destroy()

Create/Open: sem_open() → Use: sem_wait() / sem_post() → Close: sem_close() → Remove: sem_unlink("/semApex")

Blocking vs Non-Blocking

- sem_wait():
  - blocks until counter > 0
- sem_trywait():
  - Returns immediately
  - Fails with EAGAIN if count = 0
- Useful for:
  - Real-time / non-blocking tasks
  - Avoiding potential deadlocks

Initialization in Shared Memory (Inter-Process)

Create shared memory → Place sem_t inside shared region → Call sem_init(&sem, 1, initial_value); → All processes map same region and use semaphore

Ordering Constraints

Semaphores guarantee:	Do NOT guarantee:
• atomic decrement/increment • correct wake-up order (FIFO-ish, implementation-dependent) • protection of critical sections	• fairness • priority scheduling • starvation prevention

When to Use a Semaphore

- Sync between threads or processes
- Producer–consumer buffering
- Resource pool management
- Wake-up signaling
- Guarding critical regions
- Rate limiting (e.g., max N tasks allowed)

When NOT to Use Semaphores

- To protect complex data requiring ownership semantics → use mutex
- For waking specific processes → use condition variables
- For message-based communication → use MQ or pipes
- For large data transfer → use shared memory + semaphores

Deadlocks & Pitfalls

Avoid:

- Taking semaphores in different orders
- Forgetting sem_post()
- Waiting twice without posting
- Misplacing semaphores inside unmapped memory
- Using named semaphores without unlinking → leftover objects

Debugging & Observing Semaphores

- Check /dev/shm for named semaphores (Linux)
- Use strace to monitor sem_wait() / sem_post()
- Check permissions with ls -l /dev/shm/sem.
- Ensure proper unlink on exit

Summary

- Semaphores synchronize processes/threads using a counter
- Binary → mutual exclusion
- Counting → resource management
- POSIX APIs: sem_init, sem_open, sem_wait, sem_post
- Must use carefully to avoid deadlocks
- Essential in embedded systems for safe concurrency

Scenario: Parking Lot Capacity Controller (1/3)

- You're simulating a mall parking lot with limited slots (N).
- Each arriving car is a thread that tries to enter, parks for a while, and leaves.
- A counting semaphore enforces capacity so that at most N cars are parked simultaneously.
- A binary semaphore (or mutex) protects shared logs to prevent interleaved prints.

Scenario: Parking Lot Capacity Controller (2/3)

PART 1 — Core Capacity Control

- Inputs (CLI): N_slots (e.g., 5), N_cars (e.g., 20).
- Initialize counting semaphore to N_slots.
- Spawn N_cars threads; each thread:
  - Prints Car <id> arriving
  - sem_wait(&slots) → blocks if full
  - Prints Car <id> parked
  - Sleeps 1–3 sec (random)
  - Prints Car <id> leaving
  - sem_post(&slots)
  - Use a mutex/binary semaphore around printf to avoid mixing lines.

Scenario: Parking Lot Capacity Controller (3/3)

PART 2 — Arrival Patterns & Metrics

- Randomize inter-arrival times (0–500 ms).
- Collect metrics: peak waiters, avg wait time before getting a slot.
- Print a final summary table.